engineering proceedings

MDPI

Proceeding Paper

Towards Optimal Scheduling of Intermittent Water Supply Systems Incorporating Consumer Behavior †

Sriman Pankaj Boindala [1], Gopinathan R. Abhijith [2], K. Ihjas [3] and Avi Ostfeld [1,*]

[1] Faculty of Civil and Environmental Engineering, Technion—Israel Institute of Technology, Haifa 32000, Israel; srimanpankaj@gmail.com
[2] Department of Civil Engineering, Indian Institute of Technology Kanpur, Kanpur 208016, Uttar Pradesh, India; abhijith@iitk.ac.in
[3] Scientist B, Ecology & Environment Research Group, KSCSTE—Centre for Water Resources Development and Management (CWRDM), Kozhikode 673571, Kerala, India; ihjas@cwrdm.org
* Correspondence: ostfeld@cv.technion.ac.il
† Presented at the 3rd International Joint Conference on Water Distribution Systems Analysis & Computing and Control for the Water Industry (WDSA/CCWI 2024), Ferrara, Italy, 1–4 July 2024.

Abstract: Intermittent water supply (IWS) systems, originally intended for continuous supply, have been compelled to adopt intermittent supply due to factors such as water scarcity, financial limitations, ineffective operational tactics, unexpected increases in demand, and infrastructure deterioration. In response, consumers have adapted by employing flexible behaviors and utilizing storage tanks to manage water during non-supply periods. This study aims to present a methodology for devising an optimal schedule for intermittent operations, prioritizing consumer equity. The framework is tailored to a real-world intermittent network in rural South India, accounting for practical constraints and fluctuations in demand. This article only shows the preliminary analysis of the system; the development of the optimization framework is still a work in progress.

Keywords: intermittent water supply; equity; rural water distribution system

check for updates

Citation: Boindala, S.P.; Abhijith, G.R.; Ihjas, K.; Ostfeld, A. Towards Optimal Scheduling of Intermittent Water Supply Systems Incorporating Consumer Behavior. *Eng. Proc.* **2024**, *69*, 168. https://doi.org/10.3390/engproc2024069168

Academic Editors: Stefano Alvisi, Marco Franchini, Valentina Marsili and Filippo Mazzoni

Published: 25 September 2024

Copyright: © 2024 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Water supply systems (WSSs) are complex infrastructures essential in every community designed to process, store, and supply potable high-quality water at required pressures [1]. Meeting these objectives is difficult for many water utilities worldwide [2]. These water utilities operate intermittently due to factors such as water scarcity, power shortages, insufficient funds to expand network capacity, ineffective operation strategies, and deteriorated infrastructure, causing leaks. This means they frequently operate for less than 24 h a day [3]. The persistence, if not the proliferation, of intermittent water supply (IWS) networks is expected due to the un-projected increase in population growth, urban migration, and the impact of climate change on water resources [4].

Hydraulic modeling software tools like EPANET 2.2 were developed to model continuous water supply (CWS) networks, but regrettably, there is a lack of options for modeling the distinctive features of intermittent water supply (IWS) [3,5]. To address this limitation, researchers have modified existing software or developed their own to model intermittent water supply (IWS) [3].

2. Modeling Intermittent Water Supply

In typical practice, IWS networks are designed for continuous operation but are operated intermittently. To meet water demands under such erratic WDS operations, consumers generally practice flexible consumption behaviors that involve withdrawing the available water during shorter supply periods and storing it in intermediate storage facilities [5]. Consequently, flow rates in IWS often surpass the designed capacity, leading

to decreased pressures [4] and impacting consumer withdrawals. Pressure-driven analysis (PDA), initially devised to simulate consumer withdrawals during unexpected pressure deficits in CWS, has been frequently adapted to model IWS networks, which frequently experience pressure deficiencies.

Abhijith et al. (2023) [6] introduced an IWS modeling tool called EPyT-IWS, utilizing the open-source EPANET2.2 python toolkit (EPyT) [7]. This modeling approach aims to integrate the hydraulic analysis of IWS using EPANET-2.2 (EPyT) with the behavior of domestic storage tanks and consumer withdrawals from these tanks, enabling a more accurate assessment of equity and demand satisfaction. It integrates the modeling ecosystem of EPANET with an independent hydraulic solver. At each time step, the EPANET solver is applied to solve the part of the hydraulic model initialized as input data files for EPANET in (.inp format), and the independent hydraulic solver is employed to solve the remaining part, autogenerated by the EPyT-IWS algorithm. For the current study, this modeling approach is used to analyze the system's demand satisfaction and operation.

3. Case Study

This study focuses on a rural intermittent water supply system in Northern Kerala, India, serving as a relevant case study. The network's layout is depicted in Figure 1. Water is sourced from a meticulously constructed dug well, 6.1 m deep and 6 m wide, with a static water level of 3.45 m, emphasizing reliance on groundwater. Pumped efficiently to an overhead tank positioned 180 m above ground level, this tank serves as a vital distribution point for households averaging 120 m in elevation. Despite yielding approximately 187 cubic meters of water daily, prudent management suggests a withdrawal rate of 90,000 lpd to maintain recharge and prevent depletion. Design considerations accommodate an average allocation of 60 lpcd per consumer, with a carefully planned pumping schedule pumping 18,000 lph for 5 h, followed by a 2.5 h cycle and an 8 h rest period to replenish the source, ensuring system reliability.

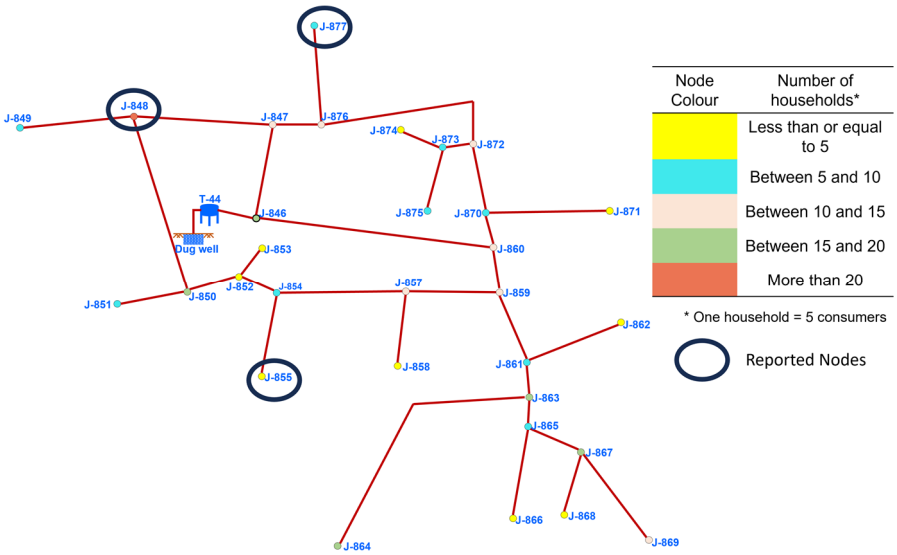

Figure 1. Case Study Network detailing the number of household distribution within the system.

Due to the increase in the living standards of the people, the water demand has increased to 135 lpcd. As this creates a heavier burden on the system, the supply has changed to intermittent supply to cater to increased demands. It is assumed that the consumers have installed 1000 L capacity tanks to store the water, changing the hydraulics of the system. From direct observation, the consumers were found practicing Type-B behavior in meeting demands from the intermittently operated WSS.

4. Preliminary Analysis

For analyzing this case study, the dug-well source is replaced with a source node with a base demand of '-5' liters per second to account for the pumping capacity of the well and a 24 h-long demand pattern, signifying 5 h pumping with a minimum 8 h interval. The primary analysis was performed using PDA and EPANET-2.2. The analysis results show a continuous high-pressure supply of water to consumers. Thus, purely based on the EPANET-based PDA analysis, we may claim that the system can satisfy the demand of 60 L per day(lpcd). The system is then subjected to an increased demand of 135 lpcd and analyzed using EPANET-based PDA. The source tank pressures are reported for both 60 lpcd and 135 lpcd in Figure 2. The demand deficit of selected nodes is reported in Figure 3.

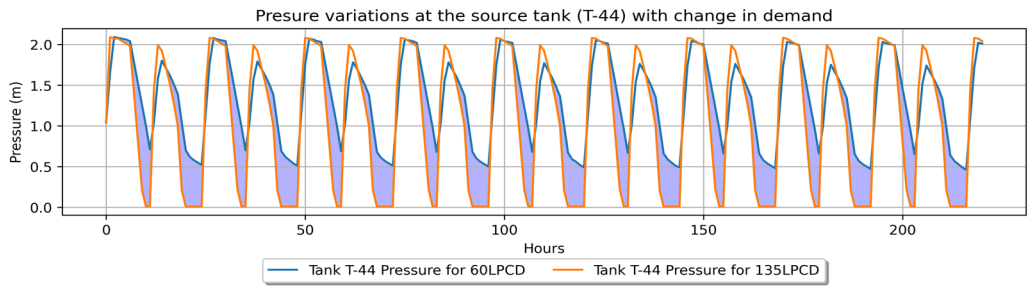

Figure 2. Pressure variations in the source tank with change in demand from 60 lpcd to 135 lpcd. The purple parts depict the deviation of the pressure in the tank.

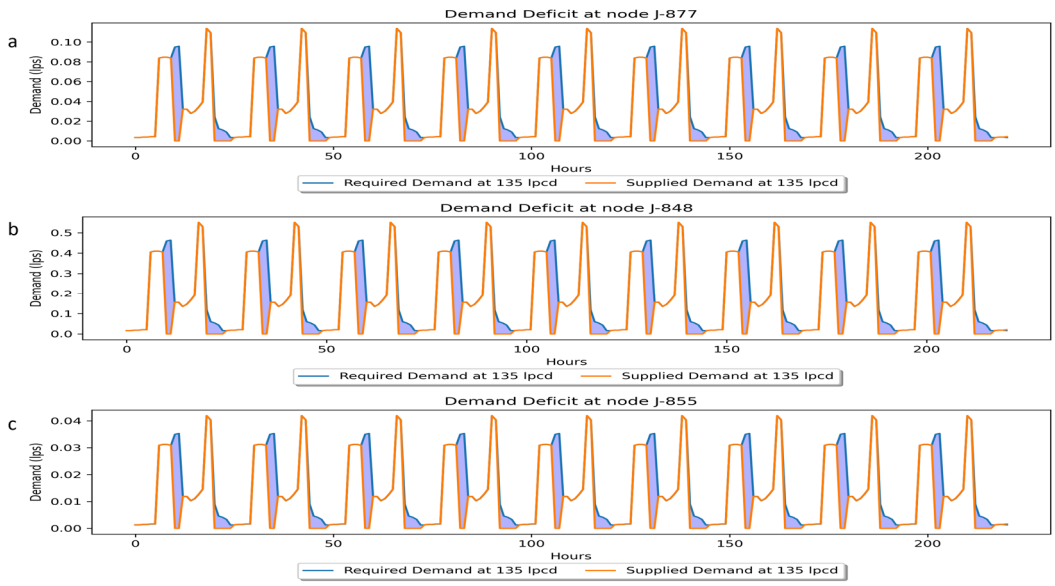

Figure 3. Demand deficit for a few nodes in the system was obtained using EPANET PDA simulations. The sub-figures are for selected nodes (J-877 (**a**), J-848 (**b**), J-855 (**c**)) from the system.

To better understand the hydraulic changes in the system due to the intermediate consumer storage and consumption patterns, the system needs to be modeled with the EPyT-IWS tool [6], employing a pressure-dependent volume-driven approach. The EPyT-IWS simulations are expected to shed light on the dynamics of the consumer response to unreliable and unreliable water supply routines. The EPyT-IWS analysis is currently ongoing.

After the analysis, an optimization problem will be solved with the decision variable to be the source node demand pattern subjected to practical constraints of the dug well reachability. The objective function is to maximize the equity of the system, which is obtained through the demand deficit results obtained from the EPyT-IWS analysis. The

optimization model would be a simulation-optimization framework combining the EPyT-IWS and cuckoo search meta-heuristic algorithm.

5. Conclusions and Future Work

Currently, the work is still in progress, and the results reported in this article are from the preliminary analysis of the real rural water distribution system of India. The network is constructed to deliver high-pressure water to its consumers. The system was originally designed to deliver 60 lpcd and can supply this demand continuously. Due to the increase in consumer demand and domestic storage tanks, the system is forced to operate intermittently. The final goal of this study is to propose a simulation-optimization framework that utilizes the advantages of EPyT-IWS for IWS simulations to obtain the demand deficit of the consumer. The optimization model will provide an optimal operation schedule of the system to maximize the demand satisfaction of the consumers. This would be the future work.

Author Contributions: Conceptualization, S.P.B., G.R.A., K.I. and A.O.; methodology, S.P.B., G.R.A. and A.O.; writing—Original draft preparation S.P.B.; writing—review and editing, G.R.A. and K.I.; supervision, A.O. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Data can be shared upon request to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Ostfeld, A.; Kogan, D.; Shamir, U. Reliability simulation of water distribution systems—Single and multi-quality. *Urban Water* **2002**, *4*, 53–61. [CrossRef]
2. Galaitsi, S.; Russell, R.; Bishara, A.; Durant, J.L.; Bogle, J.; Huber-Lee, A. Intermittent domestic water supply: A critical review and analysis of causal-consequential pathways. *Water* **2016**, *8*, 274. [CrossRef]
3. Sarisen, D.; Koukoravas, V.; Farmani, R.; Kapelan, Z.; Memon, F.A. Review of hydraulic modeling approaches for intermittent water supply systems. *AQUA—Water Infrastruct. Ecosyst. Soc.* **2022**, *71*, 1291–1310.
4. Vairavamoorthy, K.; Gorantiwar, S.D.; Pathirana, A. Managing urban water supplies in developing countries—Climate change and water scarcity scenarios. *Phys. Chem. Earth Parts A/B/C* **2008**, *33*, 330–339. [CrossRef]
5. Abhijith, G.R.; Naidu, M.N.; Boindala, S.P.; Vasan, A.; Ostfeld, A. Analyzing the role of consumer behavior in coping with intermittent supply in water distribution systems. *J. Hydroinform.* **2023**, *25*, 1766–1787. [CrossRef]
6. Abdelazeem, O.; Meyer, D.D. How to model an intermittent water supply: Comparing modeling choices and their impact on inequality. *J. Water Resour. Plan. Manag.* **2024**, *150*, 04023071. [CrossRef]
7. Kyriakou, M.S.; Demetriades, M.; Vrachimis, S.G.; Eliades, D.G.; Polycarpou, M.M. Epyt: An epanet-python toolkit for smart water network simulations. *J. Open Source Softw.* **2023**, *8*, 5947. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.